ICTC 2026 — POSTER PRESENTATION

INTERNATIONAL CONFERENCE ON ICT CONVERGENCE

Stress Prediction Using Personalized Pre-Sleep Lifelog Features

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Research Overview

BACKGROUND · DATASET · MODEL & VALIDATION

Stress is closely linked to sleep quality and mental health, and smartphones and wearables now enable continuous multimodal lifelog sensing. Yet most existing approaches rely on population-level models or daily-aggregated features and overlook each person's own behavioral baseline — **a sensor value should be read relative to a person's usual pattern and recent state, not by its absolute value alone.**

RQ1. Do questionnaire and sleep-history features improve prediction beyond raw behavioral and environmental features?

RQ2. Do personalized Z-score features provide additional predictive information?

RQ3. Does the proposed model generalize to held-out participants with available personal history?

DATASET

Smartphone and wearable lifelog data were used to predict pre-sleep stress (**Q3**). The counts represent successive stages of the analysis pipeline.

10 Participants

↓

450 Participant-days

↓ ≥5 days of prior history

400 Eligible Observations

↓ pooled across outer folds

217 Outer-test Predictions

FEATURE ROLES (93 CANDIDATES)

Raw (36) — absolute pre-sleep behavior and environment values (screen use, light, activity, movement), not adjusted for the person.

Personalized Z-score (34) — the same signals as a deviation from each person's own typical baseline.

History (20) — previous Q1–Q3/S1–S4 self-reports and recent averages, capturing a person's typical state.

Availability (3) — flags for missing or incomplete sensor data.

MODEL & VALIDATION

Earlier observations were used for training and later observations for testing. Nested LOSO evaluated performance on held-out participants when their personal history was available.

Regularized Logistic Regression
L2 or Elastic Net

↓

Nested Chronological Validation

↓

5,000 Bootstrap Resamples

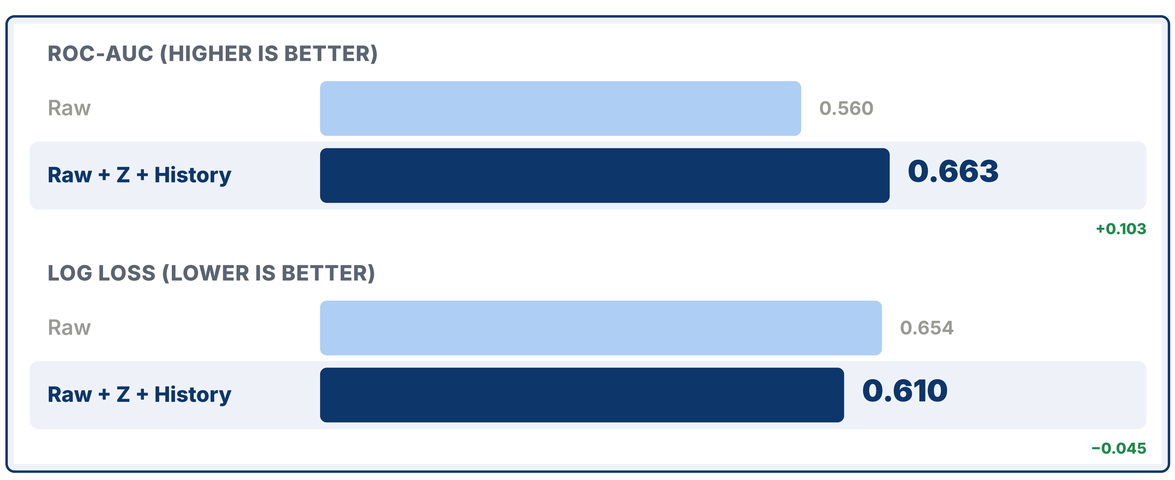
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Nested Leave-One-User-Out

Results A — Chronological Prediction Performance

PRIMARY RESULT — RQ1 & RQ2

Purpose. Does adding personalized Z-score and history features improve prediction beyond raw features alone?



95% CI excludes zero for both — **statistically significant** (ROC-AUC 0.017–0.200; Log Loss −0.078 to −0.006)

Also improved: **PR-AUC** *0.736→0.787* · **Bal. Accuracy** *0.511→0.615* · **Specificity** *0.083→0.486* (not significant)

Interpretation. Personalized and historical information drove a statistically significant improvement in the model's primary prediction measures.

Results B — Ablation Analysis

RQ1 & RQ2

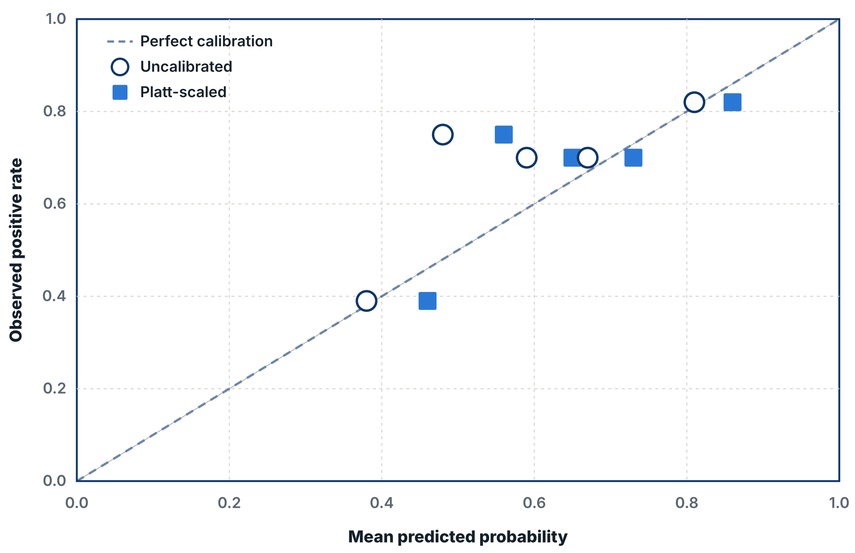
Purpose. This ablation isolates which feature group drives the improvement seen in Results A, by comparing four feature combinations directly rather than assuming more features automatically help.

Result. From Raw alone (ROC-AUC **0.560**), personalized Z-score *alone* slightly lowered ROC-AUC to **0.542** — by itself it did not help. History instead raised ROC-AUC to **0.651**, the largest single jump; combining all three reached the highest ROC-AUC, **0.663**, with Z-score adding a further, smaller gain on top of History.



Interpretation. **History features provided the largest predictive gain, while Z-score features provided complementary information; Z-score alone did not improve ROC-AUC.**

Results C+D — Calibration & Sensitivity Analysis



C. CALIBRATION

Log Loss **0.610 → 0.593** · Brier Score **0.212 → 0.203** · ROC-AUC **0.663 → 0.669**

Interpretation. Platt scaling improved both probability-calibration measures, while ROC-AUC barely moved — the model already ranked predictions well; calibration mainly made its confidence values more trustworthy.

D. SENSITIVITY ANALYSIS

Class weighting raised Specificity (**0.472→0.556**) but worsened Balanced Accuracy and Log Loss, so the unweighted model was kept. ROC-AUC reached **0.700** with 7-day history vs. ≈**0.661** for 3-/5-day, but the 7-day figure used a smaller, different sample — not evidence of an optimal window.

Results E+F — Stable Features & Cross-User Generalization

E. STABLE FEATURE ANALYSIS

Previous Q3	+0.317
Previous Q1	+0.213
Light change	+0.206
Speed change Z	−0.124

Previous Q3 was the strongest, most stable positive predictor across folds — coefficients reflect predictive association, not causation.

F. CROSS-USER GENERALIZATION

Model	Pooled AUC	Mean User AUC
Raw	0.381	0.494
Raw + Z + History	**0.643**	**0.619**

ROC-AUC improved for 8 of 10 held-out participants.

Discrimination improved with personal history available — not yet validated for a brand-new user with no history at all.

KEY TAKEAWAY

Recent personal history provided the largest predictive gain, while personalized behavioral deviations added complementary information.

Cross-user performance also improved when personal history was available, although further validation with larger populations is needed.

Conclusion

The Raw+Z+History model improved ROC-AUC from **0.560 to 0.663** and reduced Log Loss from **0.654 to 0.610**, with History features driving **most** of that gain and Z-score features adding a **complementary** boost. Cross-user discrimination also improved for held-out participants when their personal history was available.

Limitations & Future Work

The evaluation drew on only **10 participants** with incomplete sensor coverage, and the high feature-to-sample ratio (**93** features, **400** observations) raises overfitting risk, with performance for a brand-new user not guaranteed without history. Future work: larger, diverse groups; missing-data-robust methods; real-world wellness testing.

References

[1] S. Chung et al., "Real-world multimodal lifelog dataset for human behavior study," *ETRI Journal*, vol. 44, no. 3, pp. 426–437, 2022.

[2] A. Tazarv et al., "Personalized Stress Monitoring using Wearable Sensors in Everyday Settings," *EMBC*, pp. 7332–7335, 2021.

Acknowledgement. The authors thank DACON and ETRI for providing the lifelog dataset used in this study.